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Poverty Reduction or Relocation?

Long-Run Neighborhood Change Following Place-Based Policy

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Public Policy in Economic Development Faces Important Tradeoffs

- Should public policy invest in people or places?
- Place-based policies rely on private businesses to generate economic activity in distressed areas.
- Are policies reducing poverty or simply relocating it?

Main Research Question: How do place-based economic policies impact neighborhood composition, migration, and local housing costs over the medium and long-term?

What is Gentrification?

Defining Gentrification - is it a good or a bad thing?

gentrification /jĕn″trə-fĭ-kā′shən/

noun

1. The restoration and upgrading of deteriorated urban property by middle-class or affluent people, often resulting in displacement of lower-income people.
2. The process of **renewal** and **rebuilding** accompanying the **influx** of **middle class** or **affluent** people into **deteriorating** areas that often **displaces** earlier usually **poorer residents**.
3. The restoration of run-down urban areas by the middle class (resulting in the displacement of low-income residents).

The American Heritage® Dictionary of the English Language, 5th Edition • More at Wordnik

Motivation: Neighborhood conditions have lasting impacts on residents

- Consequences of economic distress
  - Communities differ in income inequality, school quality, and social capital (Chetty et al. 2014)
  - Living in struggling areas linked to poor employment, family instability, and substance abuse (Austin et. al 2018)
  - Intergenerational mobility depends on local conditions (Chetty et al. 2014)

Motivation: the government continues to use place-based policies

- Large federal investment in place-based policies
  - **Empowerment Zone** and Enterprise Community Program (1994, 1998, 2001)
  - New Markets Tax Credits (2002)
  - Opportunity Zones (2018; made permanent in 2025)

Literature: what previous research tells us

- Evidence on Empowerment Zones is mixed (Oakley and Tsao 2006, Hanson 2009)
- Some studies find modest local improvements (Rich 2010, Krupka and Noonan 2009, Reynolds and Rohlin 2014, Smith 2015)
- Broader neighborhood change remains limited (Rich 2010, Krupka and Noonan 2009)
- Round 1 raised employment and wages (Busso, Gregory, and Kline 2013)
- Long-run effects on original residents remain unclear

This paper

- **Question 1:** Do Empowerment Zones improve economic outcomes in distressed neighborhoods?
- **Question 2:** Do gains reach incumbent residents or reflect neighborhood turnover?
- **Context:** Federal place-based economic policy
  - Three designation rounds: 1994, 1998, 2001
  - Employer wage tax credits and grants
- **Method:** Matched difference-in-differences and event studies
- **Data:** Decennial Census, ACS, LBD, Economic Census, and FHFA housing data from 1989–2023

Preview of findings: did the treated neighborhoods gentrify?

- Gains in employment 5 years post-designation.
- Long-run compositional changes may mask benefits to residents.
- Residents become younger, less educated, and less white.
- Rent and home prices stay flat overall despite upward pressure from rounds 1 and 3.
- Increase in share of low-income residents and renters.

No evidence of gentrification via displacement of low-income residents: flat rents and a rising low-income share are consistent with slack housing, not capitalization.

Agenda

1. Motivation
2. Background
3. Conceptual Framework
4. Data
5. Empirical strategy
6. Results
7. Next steps

The Empowerment Zone Program

Innovative place-based policy: the federal Empowerment Zone program

Who

Census tracts meeting minimal poverty and unemployment rates.

Communities applied to be designated through competitive process.

When

Three designation rounds: 1994, 1998, and 2001.

Treatment started following year
Expired at end of 2025

What

Wage credits of up to 20 percent of the first $15,000 in wages.

Round 1 EZs received $100 million in Social Services Block Grant funds.

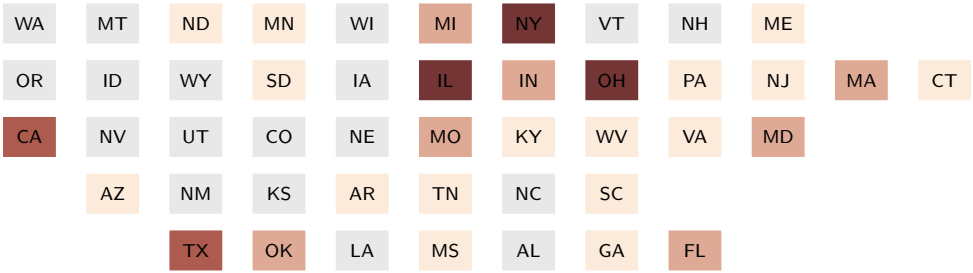
$200 million
Employment credits claimed, 1994–2000

$55 million
Credits claimed during 2000

500 co., 5,000 workers
Nearly 500 corporations and over 5,000 individuals claimed credits in 2000.

Source: Busso, Gregory, and Kline (2013); IRS (2004).

Where are Empowerment Zones?



856
Treated Census tracts

28
States represented

347
Round 1 tracts

327 & 182
Round 2 & Round 3 tracts

Source: Author's tabulation of HUD Empowerment Zone designation files.

Conceptual Framework

Conceptual Framework: Place-based benefits get priced into housing

Spatial equilibrium (Roback 1982; Moretti 2011)

Mobile households move until utility is equalized:

$$U_j = w_j - r_j + A_j = \bar{U}$$

- w_j wages, r_j housing costs, A_j amenities
- Policy raises w_j or $A_j \Rightarrow j$ more attractive
- Part of the gain is priced into higher r_j

Who captures the benefit?

- **Owners and landlords** gain when rents and values rise
- **Incumbent renters** pay higher rents, which can offset wage gains
- Incidence on residents depends on mobility and housing supply (Kline and Moretti 2014)

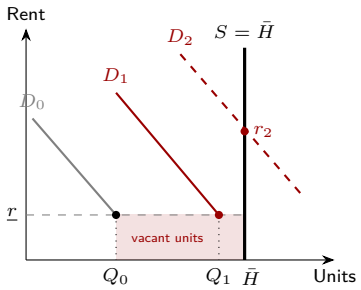
Key idea: Housing costs are the channel through which a place-based policy can turn neighborhood gains into resident turnover.

Conceptual Framework: Why Empowerment Zones could shift housing demand

- **Wage credit tied to residence:** Employers claim the 20% credit only for workers who live and work in the zone, which raises the value of living inside the boundary
- **Local labor demand:** New or expanding establishments raise employment and wages, attracting in-movers
- **Amenities and investment:** Block grants fund services, infrastructure, and safety, raising willingness to pay for the neighborhood
- **Offsetting forces:** The credit is modest (at most $3,000 per worker) and goes to employers rather than residents; take-up was weak; designation may carry stigma

Prior evidence: Busso, Gregory, and Kline (2013) find Round 1 raised resident employment and wages with little change in population or the local cost of living.

Conceptual Framework: Fixed housing stock and vacancy slack



Fixed stock $\bar{H}$

- Durable housing: distressed areas see little new construction, so the stock is fixed in the short run (Glaeser and Gyourko 2005)

Slack ($D_0 \rightarrow D_1$)

- Vacant units absorb new demand
- Occupancy rises from Q_0 to Q_1; rent stays at $\underline{r}$

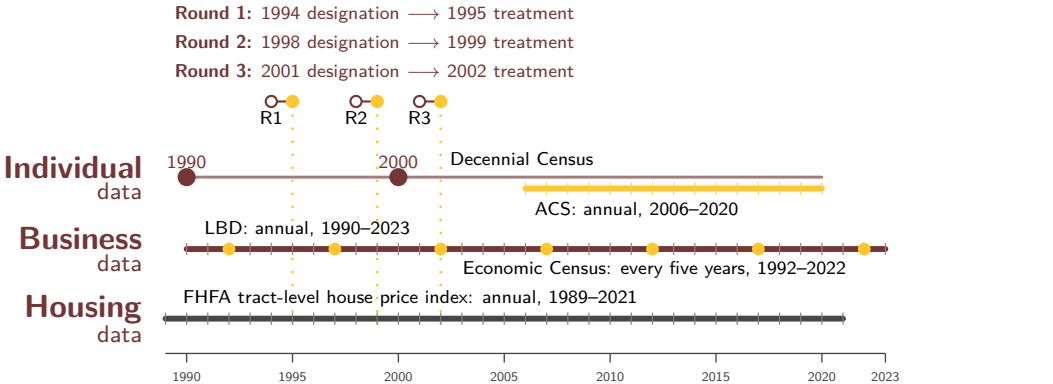
Tight ($D_1 \rightarrow D_2$)

- Once vacancies fill, rents rise to r_2
- Owners gain; renters may be displaced

Predictions: Flat rents with rising occupancy point to slack. Rising rents and values with low-income renters exiting point to capitalization.

Data

Data & Availability: National tract-year panel spanning three decades



Open circles mark designation. Gold circles and dotted lines mark effective treatment. HUD identifies more than 850 treated tracts across 28 states. Census relationship files harmonize geography to 1990 boundaries. Controls: USDA, NHGIS, BTS, and ICPSR.

Source: Decennial Census; ACS; LBD; Economic Census; FHFA; HUD.

Other data for controls come from USDA, NHGIS, BTS, and ICPSR.

Sample Restrictions and Construction: Identifying Counterfactual Tracts

- Tracts form the panel.
- People and establishments are repeated cross-sections.

Treated tracts

Original urban EZ designations

Exclude rural areas and other designations

Eligibility

Untreated urban tracts only

Poverty rate at least 20 percent

Unemployment rate at least 6.3 percent

Matched sample

Propensity-score matching selects comparison tracts

Alternate: Inverse-probability weights

Remove tracts not on common support

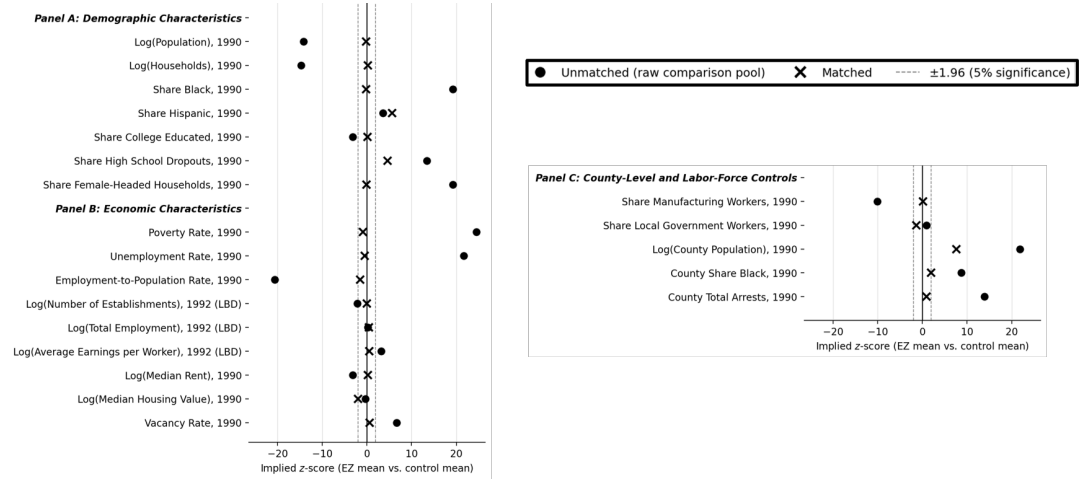
Final PSM sample: roughly 2,600 treated and matched tracts, 159,000 tract-year observations, and three designation rounds.

Balance: propensity-score matching on other characteristics

Variable	What it captures
Log(population)	Size of the tract; controls for scale and population density.
Log(households)	Number of households; complements population to capture household structure.
Percent black (or share nonwhite)	Racial composition; captures demographic structure.
Percent college (share with college degree)	Human-capital composition; local skill/education level.
Percent under 18 (share children)	Age structure and family composition; dependency and schooling demand.
Percent vacant housing	Housing market slack and neighborhood distress.
Poverty rate	General economic conditions and local prosperity.
Unemployment rate	Local labor-market slack and joblessness.
Share of Manufacturing Workers	Industrial base, agglomeration effects, and local economic structure.
Log(employment)	Baseline business activity and economic presence.
Log(average earnings per worker)	Baseline local wages / industry pay and earning potential.
County-level total crime per capita	Safety context and neighborhood quality measure.

Source: Calculations using pre-designation tract characteristics in 1990 and 1992.

Balance: propensity-score matching improves comparability of tracts



Source: Calculations using pre-designation tract characteristics in 1990 and 1992.

Empirical Strategy

Empirical Strategy: Difference-in-differences (TWFE)

Tract-year panel of EZ tracts and matched (or reweighted) control tracts:

$$Y_{jt} = \alpha_j + \gamma_t + \beta\,(Zone_j \times Post_t) + \varepsilon_{jt}$$

- Y_{jt}: tract outcome (rents, home values, income, composition shares)
- $Zone_j$: EZ tract; $Post_t$: after designation; α_j, γ_t: tract and year FE
- SEs clustered by tract, the level of treatment assignment

Identifying assumptions

- **Parallel trends:** absent designation, EZ and control tracts follow similar paths (raw business trends 1990 to 2000; event-study leads)
- **No anticipation:** announcement and effective designation were days apart

β: average change in outcome Y for EZ tracts relative to similar untreated tracts.

Empirical Strategy: Event study (TWFE)

Replace $Post_t$ with leads and lags relative to treatment year T_j:

$$Y_{jt} = \alpha_j + \gamma_t + \sum_{k \neq -1} \beta_k \left(Zone_j \times \mathbf{1}\{t - T_j = k\} \right) + \varepsilon_{jt}$$

- Leads ($k < 0$) test parallel trends; lags ($k \geq 0$) trace dynamic effects
- **Housing and business outcomes** (annual): omit $k = -1$
- **Survey outcomes**: 1989 is the only pre-period for all rounds, so it is the omitted baseline

Round	Treatment begins	1989 event time
Round 1	1995	$k = -6$
Round 2	1999	$k = -10$
Round 3	2002	$k = -12$

Survey coefficients measure the change in the EZ minus control gap relative to 1989. Rounds are estimated separately.

Empirical Strategy: Staggered designation (Borusyak, Jaravel, and Spiess 2024)

- Pooled TWFE uses "forbidden comparisons" to already-treated tracts.
- Heterogeneous effects can receive negative weights.

BJS imputation estimator (ATT)

1. Fit fixed effects on untreated observations only: $Y_{jt} = \alpha_j + \lambda_t + \varepsilon_{jt}$
2. Impute each treated tract's counterfactual and its effect:

$$\hat{Y}_{jt}(0) = \hat{\alpha}_j + \hat{\lambda}_t, \qquad \hat{\tau}_{jt} = Y_{jt} - \hat{Y}_{jt}(0)$$

3. Aggregate: $\hat{\tau}^w = \sum_{jt \in \Omega_1} w_{jt}\, \hat{\tau}_{jt}$

- Short-, medium-, long-run ATTs: equal weights within each window, estimated jointly
- Event study: $w_{jt} = \mathbf{1}\{t - T_j = h\}/|\Omega_{1,h}|$; no omitted baseline
- Most precise with annual data and for Round 3 (two survey pre-periods)

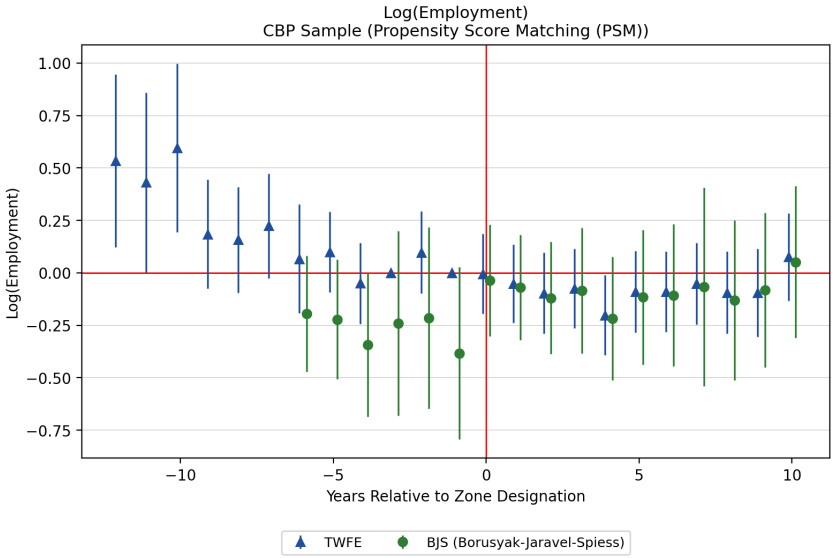
Short Run Effects: People & Businesses

Short Run Effects: Employment Gains and Rent Changes

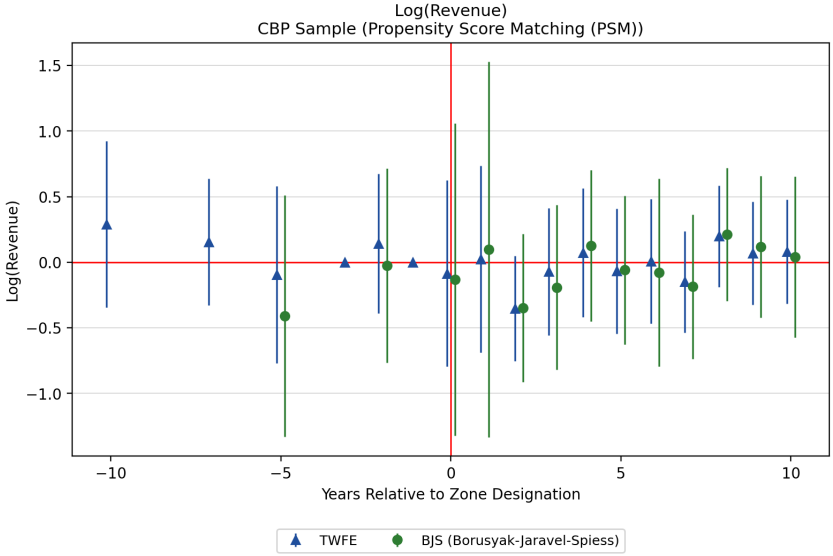
	Round 1		Round 3	
Outcome	All	Incumbent	All	Incumbent
Share Employed	0.028***	0.005	0.057**	0.039
	(0.007)	(0.008)	(0.023)	(0.027)
Share in Labor Force	−0.005	−0.018**	0.050**	0.047*
	(0.006)	(0.008)	(0.023)	(0.028)
One Month's Rent ($)	73.110***	53.540***	56.470***	120.000***
	(3.139)	(3.651)	(15.050)	(17.550)
Share Renter	−0.023***	−0.023***	−0.028	0.122***
	(0.005)	(0.007)	(0.021)	(0.026)
Share Not Moved - 1 yr (Residential Stability)	0.011**		0.057*	
	(0.005)		(0.030)	

Note: Round 2 not included due to data limitations.

Business Employment over event time



Business Revenue over event time



Results: Understanding Short Run Effects

- Increase in employment among individuals living in the tract
- No change in business revenue 10 years after designation
- No change in business employment 10 years after designation.

Composition of employees may be changing, but overall employment remains flat.

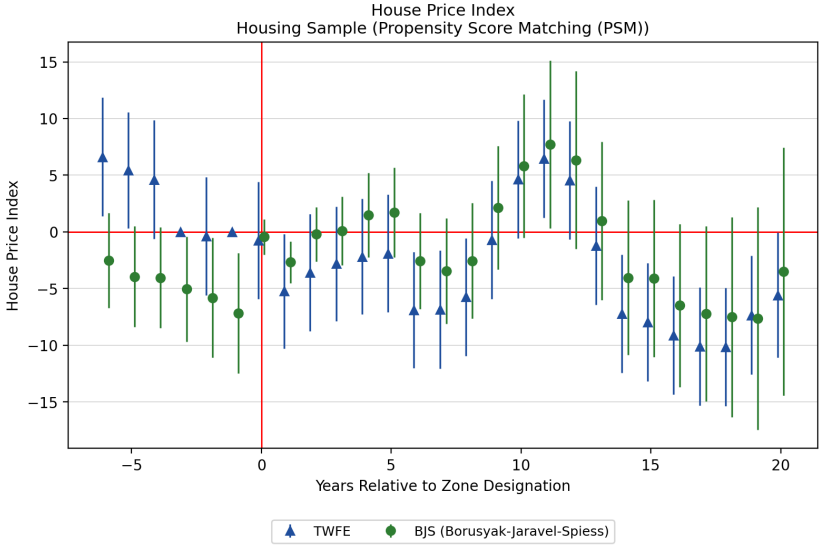
Looking for sustained effects: from short-run gains to long-run change

Who lives in EZ neighborhoods 20 years later?

- Short-run gains in employment fade over time.
- Explore long-run impacts of zone designation.
- Round-specific event studies up to 25 years post treatment.

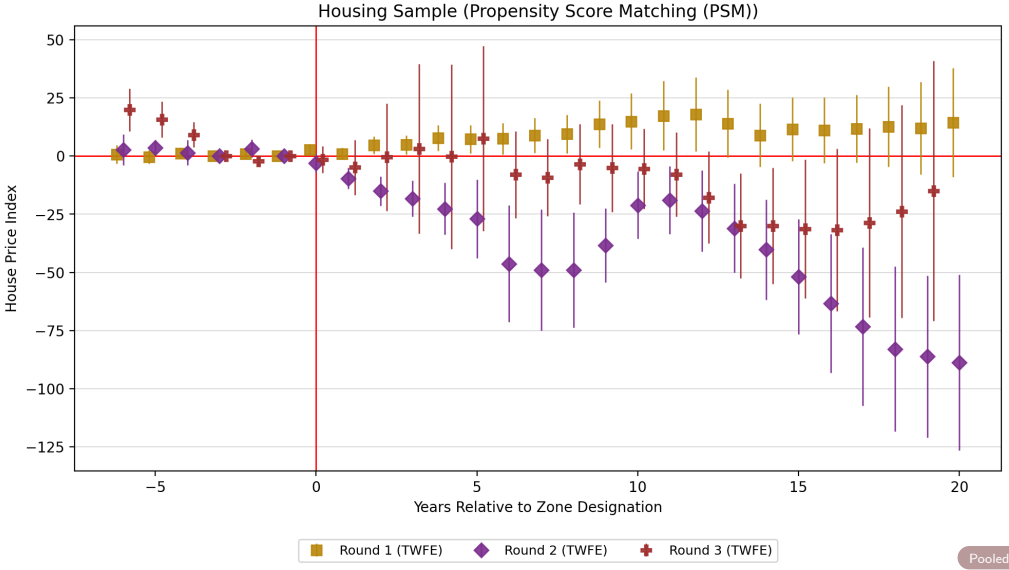
Why round-specific event studies instead of pooled like the business outcomes?

Pooled Housing Results

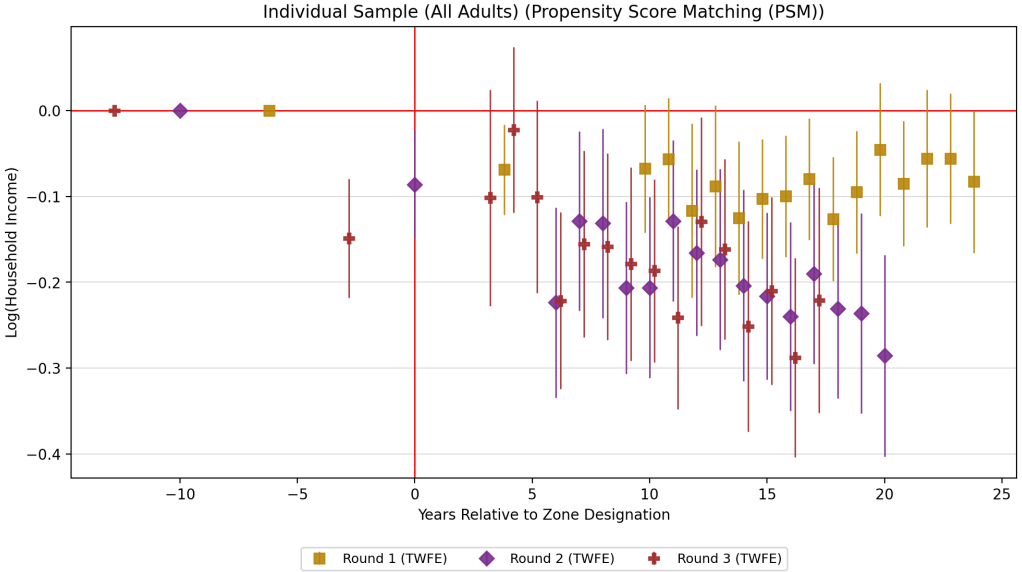


Round-specific results

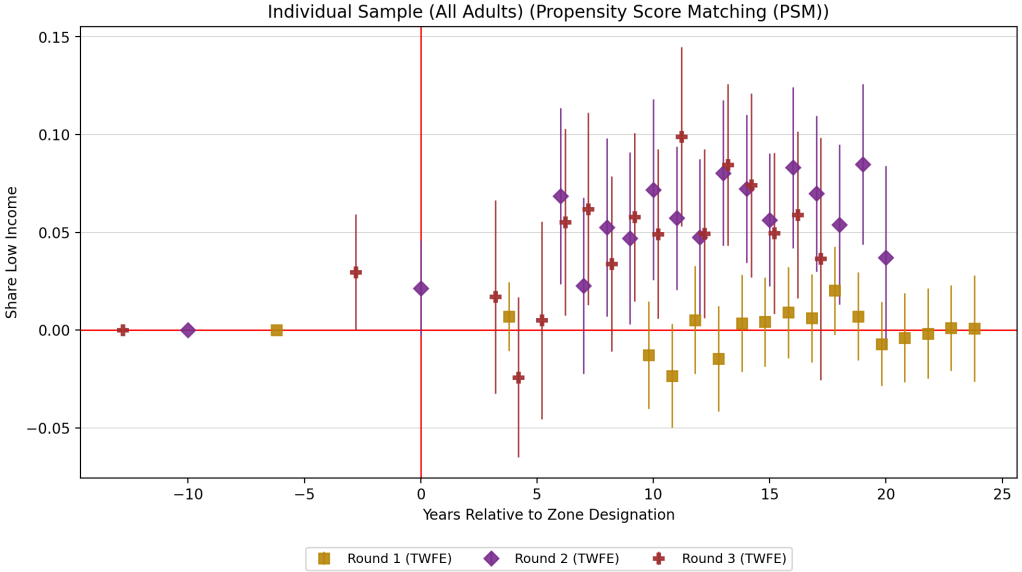
Housing values differ across designation rounds



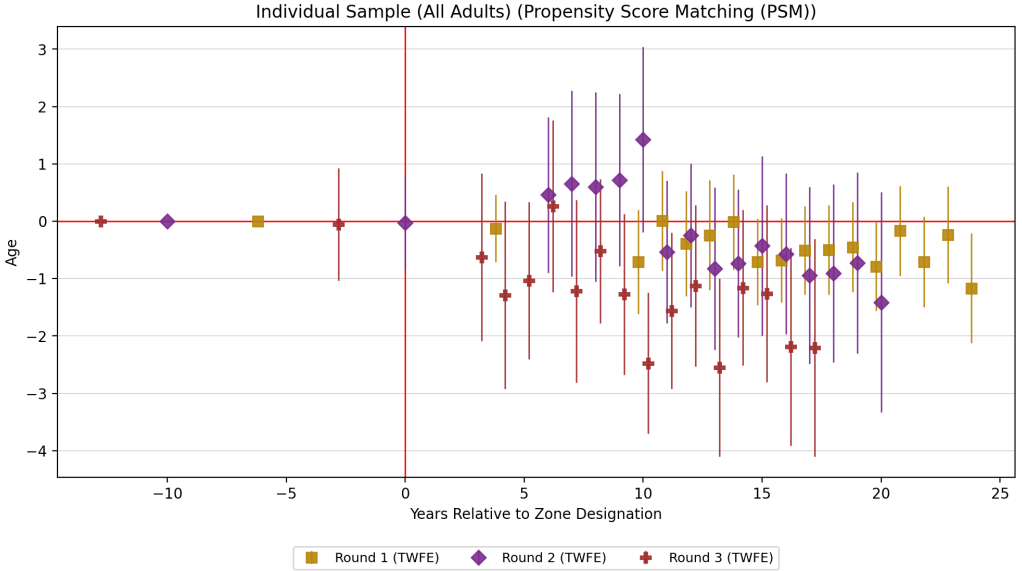
Total household income over event time



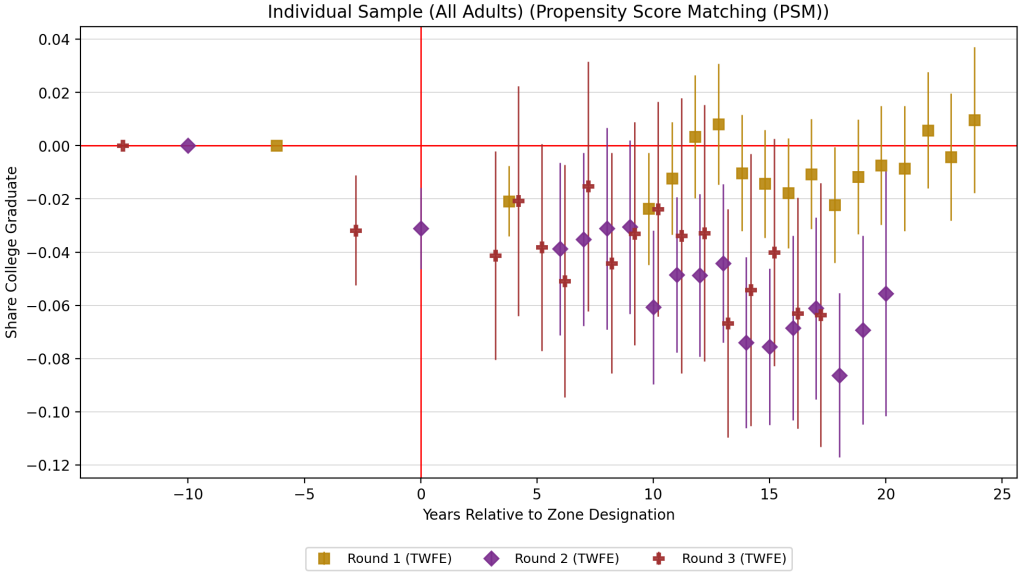
Share of Low-income Residents over event time



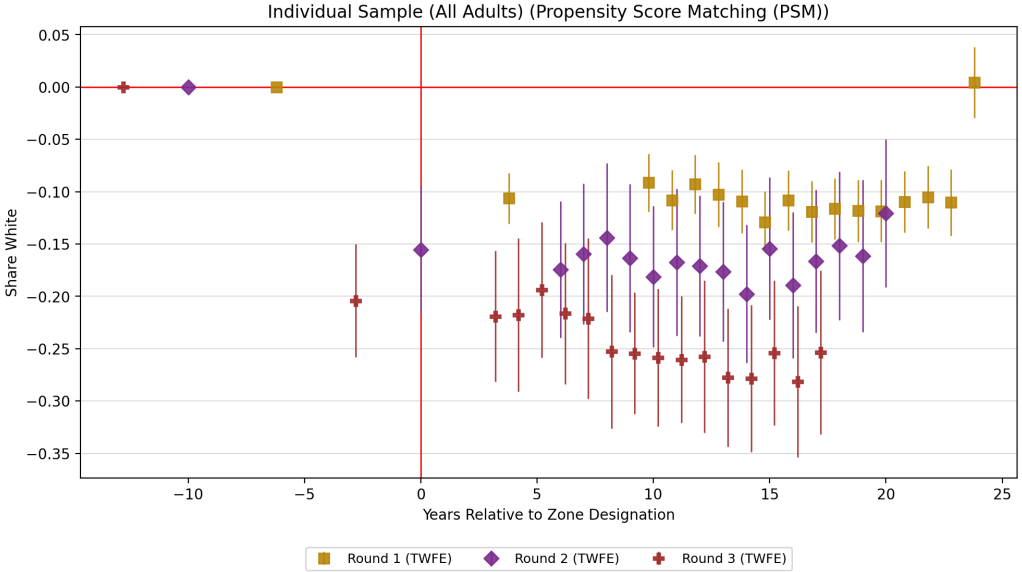
Residents become slightly younger



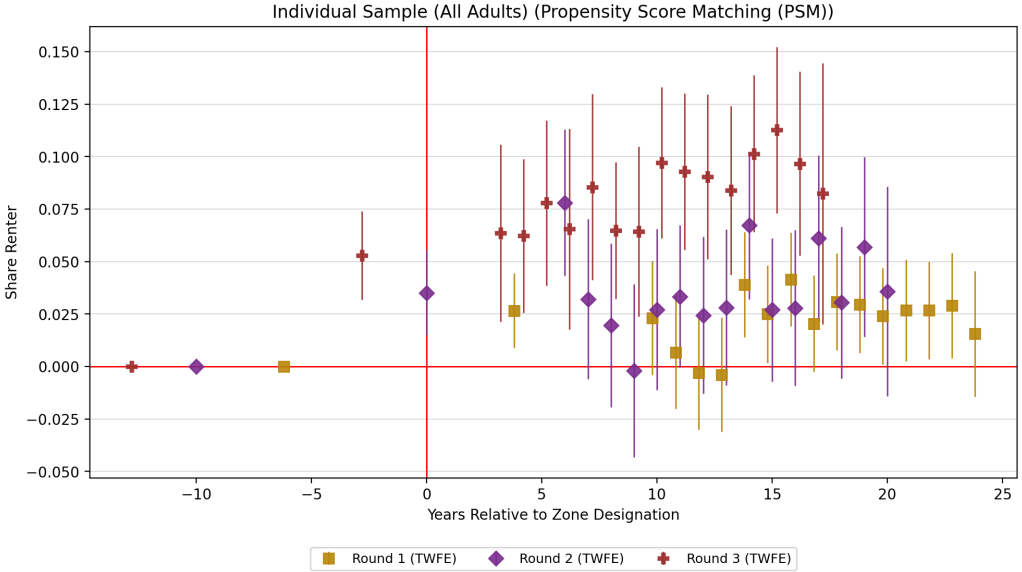
Educational composition changes over time



Racial composition shifts over event time



Home Tenure changes over event time



Conclusion: Increased residential stability without poverty reduction

What the results show

- Employment rises initially, but over time household income decreases.
- Neighborhoods become younger, less educated, and more likely to rent.
- Home values remain flat overall.
- Effects differ across designation rounds.

This evidence suggests that Empowerment Zones haven't contributed to cost-driven displacement or long-term poverty reduction.

Future Steps for the Project

Next revision

- Match Round 3 using 2000 characteristics.
- Round-specific differences, especially in housing values, merit further study.
- Use 1990 DC PIK to examine effects on individuals vs. tracts.
- Investigate mechanisms: housing supply, population, migration rates.
- Explore downward spiral of urban decline.

Tracking residents over time will inform the greater question: *who benefits from place-based policy?*

Thank you

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